Chapter 6. Estimation

Method of moments, maximum likelihood, multiparameter MLE, asymptotics of the MLE, least squares

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Setting

$X_1, \dots, X_n$ is a random sample from $f(x;\theta)$, $\theta \in \Omega \subseteq \mathbb{R}^k$. An **estimator** is a statistic $\hat{\theta}_n$ used to guess θ; it is **consistent** if $\hat{\theta}_n \xrightarrow{p} \theta$ for every $\theta \in \Omega$.

6.1 The method of moments

If $\eta = g(m_1, \dots, m_k)$ is a function of the population moments $m_r = E(X_1^r)$, the **method-of-moments estimator** (MME) replaces them by the sample moments $\hat{m}_r = \frac{1}{n}\sum X_i^r$:

$$\hat{\eta} = g(\hat{m}_1, \dots, \hat{m}_k).$$

Theorem 6.1

If $E|X_1|^k < \infty$ and g is continuous at $(m_1, \dots, m_k)$, the MME is consistent. If moreover $EX_1^{2k} < \infty$ and g is differentiable, it is asymptotically normal.

Proof. The vector of sample moments converges in probability to $(m_1, \dots, m_k)$ by the law of large numbers, and satisfies a multivariate CLT when $2k$ moments exist. Apply the continuous mapping theorem and the delta method (Chapter 5). ■

- $\hat{\sigma}^2_{\text{MME}} = \hat{m}_2 - \hat{m}_1^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$.
- The MME of a correlation, from bivariate sample moments, is the sample correlation r.
- MMEs are not unique: for Poisson(λ), $\lambda = m_1 = m_2 - m_1^2$ gives both $\bar{X}$ and $\frac{1}{n}\sum(X_i - \bar{X})^2$.

Example 6.1

For Gamma(α, β), $m_1 = \alpha\beta$ and $m_2 - m_1^2 = \alpha\beta^2$, so $\hat{\beta} = \hat{\sigma}^2/\bar{X}$ and $\hat{\alpha} = \bar{X}^2/\hat{\sigma}^2$. With $\bar{x} = 6$ and $\hat{\sigma}^2 = 12$: $\hat{\beta} = 2$, $\hat{\alpha} = 3$.

6.2 Maximum likelihood: one parameter

Definition (Likelihood and MLE)

The **likelihood** is $L(\theta) = \prod_i f(x_i;\theta)$, viewed as a function of θ for fixed data, and $l(\theta) = \log L(\theta)$. A **maximum likelihood estimator** (MLE) $\hat{\theta}$ satisfies $L(\hat{\theta}) = \max_{\theta\in\Omega} L(\theta)$.

The MLE chooses the parameter that makes the observed data most probable. When l is differentiable it solves the **likelihood equation** $\dot{l}(\theta) = 0$, but a root need not be the maximum.

Theorem 6.2

Let $\Omega = (a, b)$ and l be twice differentiable with $\ddot{l}(\theta) < 0$ on Ω and $l(\theta) \to -\infty$ at both ends. Then the likelihood equation has a unique root, and it is the MLE.

Proof. $\dot{l}$ is strictly decreasing, so it has at most one zero. Since $l \to -\infty$ at both ends and l is continuous, l attains a maximum inside Ω, where $\dot{l} = 0$. ■

Theorem 6.3 (Invariance)

If $\hat{\theta}$ is the MLE of θ and $\eta = g(\theta)$ is one-to-one, the MLE of η is $g(\hat{\theta})$. More generally, if g_1 is any function, define the MLE of $g_1(\theta)$ as $g_1(\hat{\theta})$; this agrees with the maximizer of the induced likelihood whenever g_1 can be completed to a one-to-one map (g_1, g_2).

Proof. The likelihood in terms of η is $L(g^{-1}(\eta))$, which is maximized where $g^{-1}(\eta) = \hat{\theta}$. For g_1, apply this to the one-to-one (g_1, g_2) and take the first coordinate. ■

Reparametrization can make the problem easier. For $\mathrm{Exp}(\theta)$ (mean θ), $l(\theta) = -n\log\theta - n\bar{x}/\theta$ is not concave, but in $\lambda = 1/\theta$, $l = n\log\lambda - n\bar{x}\lambda$ is strictly concave. So $\hat{\lambda} = 1/\bar{X}$ and $\hat{\theta} = \bar{X}$.

Exponential families. If $f(x;\eta) = \exp\{\eta T(x) - A(\eta) + S(x)\}$ with natural parameter η in an open interval, then $\dot{l}(\eta) = \sum T(x_i) - nA'(\eta)$ and $\ddot{l} = -nA''(\eta) = -n\,\mathrm{Var}_\eta T < 0$. Hence the MLE solves $E_\eta T(X) = \frac{1}{n}\sum T(x_i)$ (a moment equation) whenever a solution exists. Bernoulli, Poisson, exponential and normal (with one parameter known) are examples.

Example 6.2

For $U(0,\theta)$, $L(\theta) = \theta^{-n} I(\theta \geq x_{(n)})$ is decreasing on $[x_{(n)}, \infty)$, so $\hat{\theta} = X_{(n)}$. No likelihood equation is involved, and the support depends on θ; regularity conditions fail.

Numerical solution. When the equation has no closed form (e.g. logistic location), use **Newton–Raphson**: $\hat{\theta}^{(r+1)} = \hat{\theta}^{(r)} - \ddot{l}(\hat{\theta}^{(r)})^{-1}\dot{l}(\hat{\theta}^{(r)})$, started from a consistent estimate such as the MME.

6.3 Maximum likelihood: several parameters

For $\theta \in \mathbb{R}^k$ the MLE maximizes $l(\theta)$, typically by solving $\nabla l(\theta) = 0$. The multivariate analogue of Theorem 6.2: if Ω is open and convex, the Hessian $\nabla^2 l$ is negative definite on Ω, and $l \to -\infty$ at the boundary of Ω, then the root of $\nabla l = 0$ is unique and is the MLE. Invariance (Theorem 6.3) holds verbatim.

Example 6.3 (Normal)

For $N(\mu, \sigma^2)$, $\sum(x_i - \mu)^2 = \sum(x_i - \bar{x})^2 + n(\bar{x} - \mu)^2$, so l is maximized in μ at $\bar{x}$ for any σ^2. Then maximizing $-\frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum(x_i - \bar{x})^2$ gives $\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$. By invariance, the MLE of $|\mu|$ is $|\bar{X}|$ and that of σ is $\hat{\sigma}$.

Example 6.4 (Multinomial)

For counts $x_1, \ldots, x_k$ with $\sum x_j = n$, $l = \sum_j x_j \log p_j$ on the simplex. Writing $\eta_j = \log(p_j/p_k)$ turns it into an exponential family with concave log-likelihood, and the MLE is $\hat{p}_j = x_j/n$ (when all $0 < x_j < n$; boundary cases follow by direct maximization).

In multiparameter exponential families $f = \exp\{\eta' T(x) - A(\eta) + S(x)\}$, again $\nabla^2 l = -n\,\mathrm{Var}_\eta T$ is negative definite, so the MLE solves $E_\eta T = \bar{T}$.

6.4 Asymptotic properties of the MLE

Definition (Fisher information)

For one observation, the **score** is $\dot\ell(\theta;X)=\frac{\partial}{\partial\theta}\log f(X;\theta)$ and the **Fisher information** is $I(\theta)=\operatorname{Var}_\theta\dot\ell(\theta;X)$. In matrix form, $I(\theta)=\operatorname{Var}_\theta\nabla\log f(X;\theta)$.

Lemma 6.4

Under regularity conditions (common support not depending on θ, and differentiation under the integral sign allowed): (a) $E_\theta\dot\ell(\theta;X)=0$; (b) $I(\theta)=-E_\theta\ddot\ell(\theta;X)$; (c) (Kullback–Leibler) $E_\theta\log f(X;\theta')<E_\theta\log f(X;\theta)$ for $\theta'\neq\theta$, if the model is identifiable.

Proof. (a) Differentiate $\int f(x;\theta)\,dx=1$: $\int\dot\ell f\,dx=0$. (b) Differentiate again: $\int(\ddot\ell+\dot\ell^2)f\,dx=0$. (c) By Jensen with the strictly concave log, $E_\theta\log\frac{f(X;\theta')}{f(X;\theta)}<\log E_\theta\frac{f(X;\theta')}{f(X;\theta)}=\log 1=0$, unless $f(\cdot;\theta')=f(\cdot;\theta)$ a.e. ■

Theorem 6.5 (Consistency)

Suppose each l_n is strictly concave with $l_n\to-\infty$ at the boundary of Ω (as in Theorem 6.2), and $\theta'\mapsto E_\theta\log f(X;\theta')$ is finite and continuous. Then the MLE is the unique root of the likelihood equation and $\hat\theta_n\xrightarrow{p}\theta$.

Proof (one parameter). For fixed $\epsilon>0$, the law of large numbers gives $\frac1n\{l_n(\theta\pm\epsilon)-l_n(\theta)\}\xrightarrow{p}E_\theta\log\frac{f(X;\theta\pm\epsilon)}{f(X;\theta)}<0$ by Lemma 6.4(c). So with probability $\to1$, $l_n(\theta)>l_n(\theta\pm\epsilon)$. By concavity the maximizer then lies in $(\theta-\epsilon,\theta+\epsilon)$. ■

Theorem 6.6 (Asymptotic normality)

Under regularity conditions (in addition to those above: $\log f$ three times differentiable in θ with $|\dddot\ell(\theta';x)|\le M(x)$ near θ, $E_\theta M(X)<\infty$, and $0<I(\theta)<\infty$),

$$\sqrt n(\hat\theta_n-\theta)\xrightarrow{d}N(0,I(\theta)^{-1}),\qquad\text{and in } k \text{ dimensions}\quad\sqrt n(\hat\theta_n-\theta)\xrightarrow{d}N_k(\mathbf0,I(\theta)^{-1}).$$

Proof. Expand the likelihood equation around θ:

$$0=\tfrac1n\dot l_n(\hat\theta_n)=\tfrac1n\dot l_n(\theta)+\tfrac1n\ddot l_n(\theta)(\hat\theta_n-\theta)+\tfrac12\tfrac1n\dddot l_n(\theta^*)(\hat\theta_n-\theta)^2.$$

By the CLT and Lemma 6.4(a), $\frac{1}{\sqrt n}\dot l_n(\theta)\xrightarrow{d}N(0,I(\theta))$. By the LLN and Lemma 6.4(b), $-\frac1n\ddot l_n(\theta)\xrightarrow{p}I(\theta)$. The remainder is bounded by $\frac12(\frac1n\sum M(X_i))|\hat\theta_n-\theta|\xrightarrow{p}0$ by consistency. Hence

$$\sqrt n(\hat\theta_n-\theta)=\big(I(\theta)+o_p(1)\big)^{-1}\tfrac{1}{\sqrt n}\dot l_n(\theta)\xrightarrow{d}N(0,I(\theta)^{-1})$$

by Slutsky. The vector proof is identical with gradients and Hessians. ■

So $\hat\theta_n\pm z_{\alpha/2}/\sqrt{nI(\hat\theta_n)}$ is an asymptotic interval, and by the delta method $\sqrt n(g(\hat\theta_n)-g(\theta))\xrightarrow{d}N(0,g'(\theta)^2/I(\theta))$.

Example 6.5

Bernoulli: $I(p)=\frac{1}{p(1-p)}$, giving $\sqrt n(\hat p-p)\xrightarrow{d}N(0,p(1-p))$. Poisson: $I(\lambda)=1/\lambda$. Exponential with mean θ: $I(\theta)=1/\theta^2$. Normal (μ,σ^2): $I=\operatorname{diag}(1/\sigma^2,1/(2\sigma^4))$, so $\hat\mu$ and $\hat\sigma^2$ are

asymptotically independent with variances σ^2/n and $2\sigma^4/n$.

Theorem 6.7 (One-step estimator)

If $\hat{\theta}_n^{(0)}$ is $\sqrt{n}$-consistent ($\sqrt{n}(\hat{\theta}_n^{(0)} - \theta)$ has a limiting distribution), then one Newton step

$$\hat{\theta}_n^{(1)} = \hat{\theta}_n^{(0)} - \ddot{l}_n(\hat{\theta}_n^{(0)})^{-1}\dot{l}_n(\hat{\theta}_n^{(0)})$$

has the same limit distribution $N(0, I(\theta)^{-1})$ as the MLE.

Proof. Expanding $\dot{l}_n(\hat{\theta}_n^{(0)})$ around θ as above, $\sqrt{n}(\hat{\theta}_n^{(1)} - \theta) = \sqrt{n}(\hat{\theta}_n^{(0)} - \theta)\{1 - \ddot{l}_n(\theta^*)/\ddot{l}_n(\hat{\theta}_n^{(0)})\} - \ddot{l}_n(\hat{\theta}_n^{(0)})^{-1}\dot{l}_n(\theta)$. The braces are $o_p(1)$, and the last term behaves as in Theorem 6.6. ■

E.g. for the logistic location model, start at $\bar{X}$ (which is $\sqrt{n}$-consistent); one step already gives an asymptotically efficient estimator.

6.5 Least squares

Definition (Linear model)

$\mathbf{Y} = X\beta + \mathbf{e}$, with X an $n \times (p+1)$ known matrix of rank $p+1$, $E\mathbf{e} = \mathbf{0}$ and $\operatorname{Var}\mathbf{e} = \sigma^2 I_n$. The **least squares estimator** minimizes $\|\mathbf{Y} - X\beta\|^2$.

Theorem 6.8

Let $\Pi = X(X'X)^{-1}X'$. Then (a) $\Pi = \Pi' = \Pi^2$ and $\Pi X = X$ (orthogonal projection onto the column space of X); (b) $\|\mathbf{Y} - X\beta\|^2 = \|\Pi\mathbf{Y} - X\beta\|^2 + \|(I - \Pi)\mathbf{Y}\|^2$, so $X\hat{\beta} = \Pi\mathbf{Y}$ and $\hat{\beta} = (X'X)^{-1}X'\mathbf{Y}$; (c) $E\hat{\beta} = \beta$, $\operatorname{Var}\hat{\beta} = \sigma^2(X'X)^{-1}$, and $\hat{\sigma}^2 = \|\mathbf{Y} - X\hat{\beta}\|^2/(n-p-1)$ is unbiased for σ^2.

Proof. (a) Direct computation. (b) Write $\mathbf{Y} - X\beta = (\Pi\mathbf{Y} - X\beta) + (I - \Pi)\mathbf{Y}$; the cross term vanishes since $\Pi'(I - \Pi) = 0$ and $X\beta = \Pi X\beta$. The first term is 0 iff $X\beta = \Pi\mathbf{Y}$, i.e. $\beta = (X'X)^{-1}X'\mathbf{Y}$. (c) $\hat{\beta} = \beta + (X'X)^{-1}X'\mathbf{e}$. And $E\|(I - \Pi)\mathbf{e}\|^2 = \sigma^2\operatorname{tr}(I - \Pi) = \sigma^2(n - p - 1)$. ■

Under normal errors, $\hat{\beta}$ is also the MLE of β; the MLE of σ^2 divides by n instead of $n - p - 1$. Simple linear regression is the case $p = 1$: $\hat{\beta}_1 = S_{xy}/S_{xx}$, $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$.